AMERICAN PSYCHOLOGICAL ASSOCIATION

Journal of Personality and Social Psychology: Attitudes and Social Cognition

© 2025 American Psychological Association
ISSN: 0022-3514

2025, Vol. 129, No. 5, 834–849
https://doi.org/10.1037/pspa0000463

Beyond Hypothetical Trolleys: Moral Choices and Motivations in a Real-Life Sacrificial Dilemma

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Sacrificial moral dilemmas require individuals to choose between allowing harm to several people or preventing this by actively causing harm to a smaller number of people. These dilemmas have been foundational in studying how people resolve conflicts between competing moral principles, with responses traditionally interpreted as reflecting either "utilitarian" concern for the greater good or "deontological" refusal to enact harm. Existing research relies almost exclusively on hypothetical scenarios. We confronted 794 participants across two studies with real-life consequential dilemmas involving minor but genuine physical harm. Participants decided whether to allow two confederates to receive painful electroshocks or shock a third confederate and provided motivations for their decisions. Study 1 manipulated physical proximity to potential targets, while Study 2 examined how the gender composition of potential targets influenced decisions. Moreover, each participant faced this choice twice, allowing us to examine how prior outcomes influence subsequent moral decisions. Results show that responses to traditional hypothetical dilemmas moderately predict real-life behavior. Physical proximity and target gender had no significant effects on actual choices. When confronting the dilemma a second time, approximately one third of participants switched their decision, primarily to distribute harm equitably across potential targets. Analysis of participants' motivations reveals a diverse spectrum of moral considerations beyond utilitarian–deontological frameworks, including fairness concerns, responsibility avoidance, and beliefs about shared versus individual suffering. Our findings invite scholars to expand current theoretical models to better capture the nuanced ways people approach sacrificial trade-offs in consequential, real-world situations.

Statement of Limitations

Several limitations should be considered when interpreting our findings. First, our paradigm's lottery design (where only one randomly selected participant's decision determined the outcome) may have diluted participants' sense of direct responsibility for outcomes. This design element was implemented due to ethical considerations, but might have created distance from the fully consequential decision making that characterizes real-world moral choices. Second, while our use of actual electroshocks created real consequences for decisions, it remains unclear whether people would show similar response patterns when facing the life-or-death scenarios typically studied in moral psychology. Third, our sample consisted primarily of undergraduate students (predominantly female) from a single Western European university. While these sample characteristics are unlikely to affect our core theoretical conclusions, the specific proportions of choices and types of motivations observed may vary across other populations and contexts. Fourth, while participants' stated motivations provide valuable insights into their decision making, these could also, to some extent, reflect post hoc rationalizations rather than the actual psychological processes driving their choices. Finally, though our paradigm advances beyond hypothetical scenarios, it still represents a controlled laboratory environment that differs from naturalistic moral decisions in important ways, including the absence of preexisting relationships and reputational concerns.

Keywords: sacrificial dilemmas, moral decision making, ethics, utilitarianism, moral psychology

Benedek Kurdi served as action editor.
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A preprint of this article is available at https://osf.io/preprints/psyarxiv/sgznt?view_only=. This research was supported by Grant BOF22/PDO/119 awarded to Dries H. Bostyn from the "Bijzonder Onderzoeksfonds UGent". The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the article.

Dries H. Bostyn played a lead role in conceptualization, data curation, formal analysis, funding acquisition, investigation, methodology, visualization, writing–original draft, and writing–review and editing, a supporting role in resources, and an equal role in project administration. Marie-Céline Gouwy played a supporting role in investigation and writing–review and

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Would you harm a single individual if doing so saves five others? *Sacrificial dilemmas* require individuals to weigh active harm (sacrificing one or a few people) against the moral concern to minimize overall harm (saving a larger number). Moral psychologists have used countless variants of these dilemmas to study how people resolve moral conflicts (Cushman & Greene, 2012; Greene et al., 2001, 2004). Sacrificial dilemmas originate from normative philosophy (Foot, 1967; Thomson, 1976) and responses are traditionally interpreted through two ethical frameworks. Utilitarianism suggests that a morally appropriate course of action can be determined by comparing the respective outcomes of a dilemma situation and favoring whichever outcome maximizes total well-being (Mill, 2008; Sinnott-Armstrong, 2023). Deontological philosophy maintains that right and wrong are determined by absolute moral rules that should not be infringed upon; rules such as “do no harm” (Alexander & Moore, 2007; Kant, 2017). Deciding in favor of the sacrificial harm can be seen as a “utilitarian” judgment, as it minimizes the total amount of harm, whereas refusing the sacrificial harm can be construed as a “deontological” judgment, as it suggests adherence to moral rules such as the principle of not actively harming others (Conway et al., 2018).

Empirical work on this issue has mostly relied on confronting participants with *hypothetical* moral dilemmas. Consider the most famous sacrificial dilemma: The “trolley” dilemma (Foot, 1967). This dilemma describes a scenario about a runaway trolley train that is about to collide with a group of five workmen. The scenario notes that the only way to save these men is to pull a switch that diverts the trolley to a second track, where it will kill a single person instead. Participants are then asked to reflect and decide on whether one should pull the switch or not. However, contemplating such fictional moral dilemmas is profoundly different from confronting them in real life. Therefore, despite a wealth of moral dilemma research in the past decades, we are still largely in the dark about whether core insights from hypothetical dilemma research also apply when people are faced with real-life sacrificial dilemmas. To address this issue, we developed an ethically appropriate, lab based, sacrificial dilemma paradigm involving minor but real physical harm.

Contextual Factors in Moral Decision Making

Real-life moral decisions are embedded within rich contextual frameworks. People make moral decisions at varied physical and psychological distances, while confronting specific individuals with particular characteristics, as part of ongoing interactions where past choices influence future ones. While some of these factors have been examined in hypothetical dilemma research, others have received less attention. We aimed to examine their impact when people face consequential real-life moral decisions.

We first investigated physical proximity. Research has consistently shown that people tend to approve of (hypothetical) sacrificial harm when it can be accomplished indirectly from a distance, but predominantly refuse such harm when it requires up-close personal force (Cushman, 2013; Cushman et al., 2006; Hauser et al., 2007; Tasso et al., 2017). This distinction between *personal* and *impersonal* dilemmas has profoundly shaped moral psychological theories about how emotion and cognition interact in moral judgment (Greene et al., 2001, 2009). The dominant theoretical account suggests that up-close personal action triggers increased emotional processing, which leads to a higher likelihood of refusing sacrificial harm, while greater physical and psychological distance facilitates more abstract, deliberative processing that increases approval of sacrificial harm (Greene et al., 2004; Koenigs et al., 2007; Moore et al., 2011). In a first study, we therefore manipulated physical proximity, one aspect of this distinction, to test whether being physically closer to the potential victims would affect real-life moral decisions, even when the action itself (pressing a button) remained constant across conditions.

In a second study, we examined another contextual factor: the gender of potential victims. Real-life moral decisions involve specific individuals with identifiable characteristics that influence how they are perceived and valued. Yet as Hester and Gray (2020) noted, targets in moral psychology research often lack even basic identity information such as race and gender. This represents a significant disconnect between experimental paradigms and real-life moral decision making. While considerable research has examined how a decision-maker’s gender influences moral judgments (Armstrong et al., 2019; Capraro & Sippel, 2017; Friesdorf et al., 2015), fewer studies have investigated how the gender of potential victims affects sacrificial choices (Awad et al., 2018; FeldmanHall et al., 2016). In our first study (focused on physical proximity), we controlled for potential target gender effects by using same-gender confederates within each session. In our second study, we systematically varied the gender composition of potential victims (male vs. female) to examine how this factor influences real-life sacrificial decisions.

editing and an equal role in data curation and formal analysis. Elias De Craene played a supporting role in formal analysis, methodology, validation, and writing–review and editing and an equal role in investigation. Caro Vanmechelen played a supporting role in formal analysis, methodology, validation, and writing–review and editing and an equal role in investigation. Joyce Scheirlinckx played a supporting role in investigation and writing–review and editing. Tassilo T. Tissot played a supporting role in investigation and writing–review and editing. Ruben Van Severen played a supporting role in investigation and writing–review and editing. Daphne van den Bogaard played a supporting role in investigation and writing–review and editing. Milena Waterschoot played a supporting role in investigation and writing–review and editing. Fien Geenen played a supporting role in investigation and writing–review and editing. Hilde Depauw played a supporting role in investigation and writing–review and editing. Jakke Coenye played a supporting role in investigation and writing–review and editing. Juliette Taquet played a supporting role in investigation and writing–review and editing. Xinyi Xu played a supporting role in validation, visualization, and writing–review and editing. Kim Dierckx played a supporting role in investigation and writing–review and editing. Stefaan Van Damme played a supporting role in resources, supervision, and writing–review and editing. Alain Van Hiel played a supporting role in funding acquisition, project administration, supervision, and writing–review and editing. Arne Roets played a lead role in supervision, a supporting role in conceptualization, funding acquisition, methodology, and writing–original draft, and an equal role in project administration and writing–review and editing.

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Additionally, both studies incorporated a temporal dimension often absent from traditional sacrificial dilemma research: repeated decisions affecting the same individuals. The typical approach in moral dilemma research is to present participants with a series of independent scenarios, each describing a different moral dilemma. For instance, a participant might first respond to the trolley dilemma, then to a scenario about sacrificing one person to save five in a lifeboat, followed by a dilemma about harming one patient to save five others in a hospital. Each of these decisions is treated as completely separate from the others. While methodologically convenient, this strips away the temporal context of real-world moral decision making. In real life, decisions are embedded within an ongoing stream of events where past outcomes help inform what is morally appropriate in the current moment. For instance, judgments of fairness depend not only on immediate outcomes but on procedural considerations (Tyler, 2003). Similarly, studies on distributive justice show that people often strive for equality in resource allocation across time, even when this conflicts with maximizing immediate welfare (Dawes et al., 2007). In sacrificial contexts specifically, recent research suggests that people may approach repeated moral decisions differently from one-shot scenarios (Bostyn & Roets, 2022; Roets et al., 2020), with concerns regarding equality in harm allocation sometimes overriding both utilitarian and deontological concerns. By repeating the dilemma with the same victims, we examined how prior outcomes shaped subsequent decisions and whether justice considerations also emerge in sacrificial contexts.

Predictive Value of Hypothetical Choices on Real-Life Decisions

Beyond investigating how specific contextual factors impact judgment, our research aimed to address a fundamental question about moral psychology: How well do responses to traditional hypothetical dilemmas predict behavior in real, consequential situations? Sacrificial dilemmas often describe fantastical situations involving runaway trolleys or emergency organ transplants that few people would ever encounter. Critics argue that such unrealistic scenarios might fail to activate the psychological processes that drive real-world moral decisions (Bauman et al., 2014; Davis, 2012; Elster, 2011; Gigerenzer, 2008; Kneer & Hannikainen, 2022; Körner & Deutsch, 2023; Körner et al., 2019). There is some evidence to support these concerns: Bostyn et al. (2018) found that responses to hypothetical sacrificial dilemmas failed to predict behavior in a real-life sacrificial situation where participants believed mice would be harmed. However, that study involved harm to animals rather than humans.

Testing whether hypothetical responses predict behavior in a paradigm involving real human targets would provide crucial evidence about how hypothetical decisions relate to consequential moral decision making. While people rarely encounter the life-or-death situations described in typical sacrificial dilemmas, moral psychologists have nevertheless extensively relied on such high-stakes scenarios, arguing that responses to these dramatic hypotheticals reveal fundamental aspects of moral cognition that generalize to real-world decisions. The present research directly tests this assumption by examining whether established high-stakes hypothetical dilemmas predict behavior in more modest but genuinely consequential sacrificial situations that align more with the type of moral trade-offs people face in everyday life. If hypothetical responses fail to predict choices in such scenarios, this would suggest an important disconnect between traditional moral dilemma research and actual moral decision making. Such a finding would not invalidate hypothetical dilemma research (Białek et al., 2019), but would highlight the need to more carefully consider how and when findings from hypothetical dilemma research translate to real-world moral behavior.

Motivations: Utilitarian, Deontological … and Beyond?

Finally, when participants respond to sacrificial dilemmas in traditional paradigms, they are typically asked to make binary judgments or rate the appropriateness of the two options. Researchers then interpret these responses as reflecting “utilitarian” or “deontological” concerns. However, there has been considerable debate about whether these philosophical labels capture the underlying psychological processes. Critics argue that “utilitarian” choices may stem from reduced empathy rather than genuine cost–benefit reasoning (Bartels & Pizarro, 2011; Gleichgerrcht & Young, 2013; Kahane, 2012, 2015; Koenigs et al., 2007). Similarly, decisions to refuse sacrificial harm might stem not only from principled “deontological” opposition to causing harm, but from moral indifference or an unwillingness to take responsibility. Defenders of these philosophical labels have argued that while sacrificial choices might not capture the full scope of either philosophy, they do reflect the core principles: a utilitarian cost–benefit analysis for choices to commit harm and a deontological adherence to rules against causing harm for choices that refuse it (Conway et al., 2018).

While the utilitarian–deontological framework remains influential, computational moral psychology increasingly incorporates additional factors. Models based on universalization principles (Levine et al., 2020), resource–rational contractualism (Levine et al., 2024), and neural value representations (Crockett et al., 2017) suggest moral cognition involves processes beyond utilitarian calculations or deontological constraints. Nevertheless, the traditional framework continues to influence empirical interpretations, making it important to examine how well these concepts capture real-world decisions. Investigating how people explain their own choices when facing a consequential sacrificial dilemma in a real-life setting can provide a novel perspective on this debate. If choices to commit sacrificial harm reflect utilitarian reasoning, people may justify such decisions by referring to aggregate welfare. Similarly, if refusals reflect deontological concerns, these choices may be justified through appeals to moral rules against harming others. Examining real sacrificial decisions could also reveal moral considerations beyond these traditional frameworks, particularly since real consequences might prompt people to think about practical and interpersonal aspects of their choices rather than (only) abstract moral principles. Some caution is warranted when interpreting such self-reported motivations, as they might reflect post hoc rationalizations rather than the actual drivers of moral judgments (Haidt, 2001). Nevertheless, systematic patterns in these motivations could still illuminate the moral values people draw upon when explaining sacrificial dilemma decisions.

Real Moral Trade-Offs in the Laboratory

Sacrificial dilemmas are often portrayed through fantastical scenarios. We developed a novel paradigm that preserves their essential moral conflict while using real consequences: Participants

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faced a choice between allowing two confederates to receive painful electroshocks or redirecting the shock to a single confederate. This approach builds upon earlier work in moral psychology that used electric shocks to study moral cognition in study designs with consequential choices (Crockett et al., 2014; FeldmanHall et al., 2012), but represents the first implementation of a sacrificial dilemma with real physical consequences. By moving beyond hypothetical scenarios, we aimed to bridge the gap between laboratory research and the complex realities of real-life moral decision making.

Method

Participants

This article reports on two studies using the same real-life sacrificial dilemma paradigm with different contextual manipulations. Study 1 examined the effect of physical proximity (personal vs. impersonal conditions), while Study 2 investigated the influence of the gender composition of potential targets. In both studies, participants completed two stages of data collection: an online survey including hypothetical dilemmas (Stage 1) and a real-life dilemma in the lab (Stage 2).

In Study 1, a total of 485 participants started the study, with 438 completing both stages. Due to data linking issues, data are available for 434 participants. Some participants did not complete one or more questions on the article surveys used during the Stage 2 lab sessions. We retained all participants with available data and conducted analyses on a case-by-case basis depending on data completeness. Participants were sampled predominantly from the student population of a Western European university and were compensated with either course credit or financial compensation (7.5€). Participants identified as White (n = 408), mixed (n = 41), other (13), Asian (n = 9), Middle Eastern or North African (n = 6), or Black (n = 4). Most participants identified as female (371), 59 as male, and three as nonbinary. Participants had an average age of 19.65 years (SD = 3.73). No participants halted participation during the real-life dilemma experiment.

A total of 458 participants started Study 2, with 413 completing both stages. Due to technical issues with data linking procedures, complete data are available for 356 participants, who form the final sample for all analyses. Study 2 had a similar demographic make-up: White (n = 305), mixed (n = 19), other (n = 6), Asian (n = 13), Middle Eastern or North African (n = 3), and Black (n = 10). The sample included 294 female participants, 60 male participants, and one nonbinary participant. The average age was 19.04 years (SD = 5.72). One participant in Study 2 withdrew from the laboratory experiment after hearing the briefing.

Ethics

The experiments were conducted in accordance with the research protocols of Ghent University and received explicit ethical approval from the Ethics Committee (cf. institutional review board) at Faculty of Psychology and Educational Sciences (special ethical protocol:2043-043A). Due to the unique nature of our study (involving real physical harm), the ethics review process was exceptionally thorough, requiring multiple revisions over a period of 18 months and culminating in a formal hearing before the full Ethics Committee. Several design features were specifically implemented as ethical safeguards. The "lottery" element (see below, where we communicate to the participants that only one randomly selected participant's decision determined the outcome for an entire group) was implemented to minimize overall harm to the targets (i.e., confederates), while also reducing the psychological burden on participants. Similarly, the use of modest electroshocks rather than more severe consequences was aimed at striking a balance between creating genuinely consequential moral decisions and ensuring participant and the targets' welfare. The role of the targets was assumed by confederates who provided informed consent and could specify their preferred shock intensity within safe parameters.

Participants received an information letter before signing up for the experiment. The briefing in each experimental session explicitly distinguished our research from Milgram-style obedience studies and emphasized participants' right to withdraw at any time. Following the experiment, we conducted a thorough debriefing, including disclosure about a minor deception, that is that outcomes were predetermined rather than caused by participants' decisions. This was implemented in line with our Ethics protocol to minimize potential feelings of lingering guilt that participants might experience afterward if they knew their specific decision had directly caused harm to others. Participants were offered psychological assistance if needed, both immediately after participation and through follow-up contact. No participants requested follow up.

Transparency and Openness

All data, materials, and code used for both studies are available at a public repository (https://osf.io/gbm6w/; Bostyn et al., 2025). Study 1 was not preregistered and can be considered exploratory. We report how we determined our sample size, all data exclusions, all manipulations, and all measures in the study. The sample size was determined by the size of the available subject pool. A sensitivity analysis conducted based on the realized sample size indicates that the study had >80% power to detect differences in proportions of at least 0.13 and standardized odds ratios of at least 1.31.

For Study 2, we developed a preregistration after data collection but before data analysis: https://osf.io/gj7ah. This represents a "postregistration" rather than a traditional preregistration, as it was completed after data collection. To constrain researchers' degrees of freedom, we preregistered that we would conduct the same analyses as those conducted in Study 1. Importantly, the analysis code for Study 1 was made publicly available through the OSF and time-stamped before Study 2 data was collected. Sample size for Study 2 was determined by the size of the available subject pool. A sensitivity analysis indicates that the study had >80% power to detect differences in proportions of at least 0.15 and standardized odds ratios of at least 1.35.

Procedure

The procedure for both studies followed the same two-stage design: an online survey (Stage 1) followed by a laboratory session with the real-life dilemma (Stage 2).

Stage 1: Online Survey

In Stage 1, participants responded to an online survey. In this survey, they first completed demographic information (gender, age, and ethnicity) and created a secret identifier code so that we could

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link their Stage 1 responses to their Stage 2 data. In both studies, participants completed the Oxford Utilitarianism Scale (Kahane et al., 2018), the Need for Cognition scale (Cacioppo & Petty, 1982), and the Primary Psychopathy scale (Levenson et al., 1995). In Study 2, participants additionally completed the Moral Foundation Questionnaire (Graham et al., 2009), the Behavioral Inhibition Scale subscale of the Behavioral Inhibition and Activation Scale (Carver & White, 1994), the Ambivalence Toward Men Inventory (Glick & Fiske, 1999), the Attitudes Toward Women scale (Spence & Helmreich, 1972), and the Cognitive Reflection Test (Frederick, 2005). These measures were included for exploratory reasons. Analyses involving these measures are not reported in the main text, but correlation tables of all measures are available as additional online material on the OSF (https://osf.io/gbm6w/).

Finally, participants responded to a battery of four hypothetical sacrificial dilemmas. On each dilemma, participants were asked to rate how morally appropriate they found it to commit the sacrificial harm and how morally appropriate they found it to refuse the sacrificial harm on a 5-point scale of moral appropriateness. All dilemmas are available in the additional online material on the OSF (https://osf.io/gbm6w/). Upon completion of Stage 1, participants were redirected to a calendar application where they could book a timeslot to participate for Stage 2, that is, the lab-based experiment. On average, participants participated in the lab-based experiment 8.5 days after completion of Stage 1.

Stage 2: Laboratory Setup

In both studies, participants arrived at the lab in groups of 3–11 participants. They entered the lab as a group and encountered three "targets" (labeled as A, B, and C) connected to electroshock machines (DS5 Digitimer) via electrodes attached to their hand.

All "targets" were volunteer confederates who provided informed consent and could specify their preferred shock intensity within safe parameters (220 V with amperage varying from 10 to 100 μA).

Initial Briefing. Participants were briefed by an experimenter about the nature of the experiment and asked to complete an informed consent form. A translation of the briefing is available in the additional online material on the OSF (https://osf.io/gbm6w/). The briefing referenced the online survey dilemmas and explained that participants would engage with a similar dilemma in the lab. The experimenter explained that Targets A and B would receive painful but medically safe electroshocks unless participants chose to redirect the shock to Target C instead.

Participants were informed that targets had volunteered for the experiment and that each session used a different set of targets. They were told that the seating position of each target (i.e., who was A, B, or C) had been randomly determined. Participants were told that shocks were of comparable strength for all targets. To provide participants an indication of how painful the shocks were, we told them the shocks were about as painful as the sting of an injection needle or a strong pinch. The briefing explicitly distinguished the experiment from Milgram-style obedience research and emphasized participants' right to withdraw.

Decision Procedure. Upon completion of the briefing, participants exited the lab and were asked to reenter it on an individual basis to make their decision. Once inside, participants could either walk up to a marker on the floor and back out (if they did not wish to interfere) or press a button if they wanted Target C to receive a shock rather than A and B. Participants were told that their choices would not be executed immediately. Instead, their responses would be transmitted to a nearby laptop that would randomly select one response from the group of participants. After all participants had provided a response, they all reentered the lab and the laptop selected an outcome that was subsequently executed by the experimenter. While we told participants that one response would be randomly selected, the outcome per session was actually determined in advance to ensure an equal distribution of both outcomes across experimental sessions.

Participants were not intended to interact with the targets. Although participants in the personal condition (see below) could potentially engage the targets, most did not. Occasionally, a participant would offer an apology before or after making their decision. If participants attempted to initiate conversation, targets replied that they had been instructed not to engage. Targets were instructed to maintain neutral expressions throughout the experiment, without specific instructions regarding eye contact. When shocked, targets displayed genuine reactions ranging from a twitch of the hand connected to the generator to audible expressions of discomfort ("ouch") and more pronounced movements. We deliberately chose not to standardize these reactions, allowing targets to respond naturally based on their individual sensitivities and their chosen shock intensities.

Study-Specific Manipulations. In Study 1, we manipulated physical proximity to the potential targets. Approximately half of all participants completed the *personal* variant of the dilemma set-up ($n = 226$). These participants completed the experiment in the same room as the one in which the targets were seated. The button that participants needed to press to shock the single target was attached to the left arm of that target. Accordingly, if participants wanted to commit sacrificial harm, they had to walk up to that person and press the button attached to Target C. In the *impersonal* variant, participants ($n = 212$) completed the experiment in a room with a one-way mirror, adjoining the room where the targets were seated. The button to commit sacrificial harm was located in their own room, but participants saw the confederates through the one-way mirror. Participants were informed that they could see the targets but the targets could not see them. Within each session, target gender was held constant, with target gender balanced across sessions (i.e., all targets within a session were either male or female).

In Study 2, all participants completed the personal variant of the dilemma (same room as the targets with the button attached to Target C). Instead of manipulating physical proximity, we systematically varied the gender composition of the targets. In the male-minority condition ($n = 188$), Targets A and B (who would receive shocks unless the participant intervened) were female, while Target C (who would receive a shock if the participant chose to intervene) was male. In the female-minority condition ($n = 168$), Targets A and B were male, while Target C was female.

Outcome and Motivations. After the target(s) received a shock, participants were asked to complete a paper survey. This survey probed participants about their decision and their motivation for that decision. Additionally, participants rated the moral appropriateness of both options in the dilemma, their doubt, deliberation, discomfort levels, and skepticism about the usage of real shocks. Once participants completed the survey, the dilemma was then repeated a second time. The experimenter explained that the same Targets A and B were once again exposed to being shocked and that

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they needed to decide yet again whether to shock Target C instead. The experimenter reiterated that participants were free to halt participation. Participants then completed the same decision procedure as before. Once all participants had completed the dilemma again, another outcome was selected and executed, and participants completed a second paper survey. This final survey included the same questions as before, except the question on skepticism. Additionally, it included questions probing what participants would do on a third trial, whether they had experienced the experiment as emotionally taxing, whether they preferred the current experiment over other, more standard experiments they may have participated in, and what (if anything) they had heard about the experiment before their participation.

Debriefing and Follow-Up. After all participants completed the second survey, they were debriefed about the goal of the study and the misdirection we had used (i.e., that their decisions were actually unrelated to the dilemma outcome). At the end of the debriefing, we probed whether any participants felt upset by the experiment and, following the requirements of our Institutional Ethics Committee, suggested that psychological assistance could be provided if any participants felt this was necessary. This offer was repeated in an email sent to participants after their participation. No participants asked for psychological assistance after their participation.

Statistical Analysis and Motivations

All data were analyzed in the R statistical environment (R Core Team, 2012). To categorize participants' motivations, the first and second authors jointly coded the first 100 motivations of the first and second dilemma iterations from Study 1. The first author then coded all remaining motivations and provided the second author with a codebook explaining how all categories were coded. The second author then independently coded all motivations using the codebook. To assess interrater reliability, we looked at the confusion matrices, quantity disagreement, and allocation disagreement. In both studies, the coding categories "Inaction", "No harm" and "No Responsibility" demonstrated conceptual overlap. Nevertheless, overall agreement was assessed at >.98 for the coding of T1 and T2 in both Studies 1 and 2. Differences in coding were discussed to arrive at final categorizations. The coding procedure was identical across both studies.

Results

Initial Decision

As an initial analysis, we tested whether participants' responses to the first iteration of the sacrificial dilemma differed across our two studies. A logistic regression on first-iteration choices with study (dummy-coded) as a categorical predictor revealed no effect of study on participants' decisions ($\hat{b} = 0.18$, $z = 1.23$, $p = .218$).

Proximity Manipulation

We then analyzed whether physical proximity (dummy-coded) influenced sacrificial choices in Study 1. Interestingly, although the personal variant required participants to walk up to the target and make physical contact if they wanted to redirect the shock, a logistic regression model revealed that this did not impact the likelihood of making the sacrificial decision: 58% of participants in the personal condition and 59% of participants in the impersonal condition inflicted the sacrificial harm ($\hat{b} = -0.04$, $z = -0.212$, $p = .832$). Participants' moral approval ratings corroborated these behavioral findings: A linear regression uncovered no differences for participants' approval for committing sacrificial harm, $\hat{b} = 0.00$, $t(436) = 0.01$, $p = .993$, or their approval for refusing harm across Study 1 conditions, $\hat{b} = -0.03$, $t(436) = -0.30$, $p = .762$.

Gender Manipulation

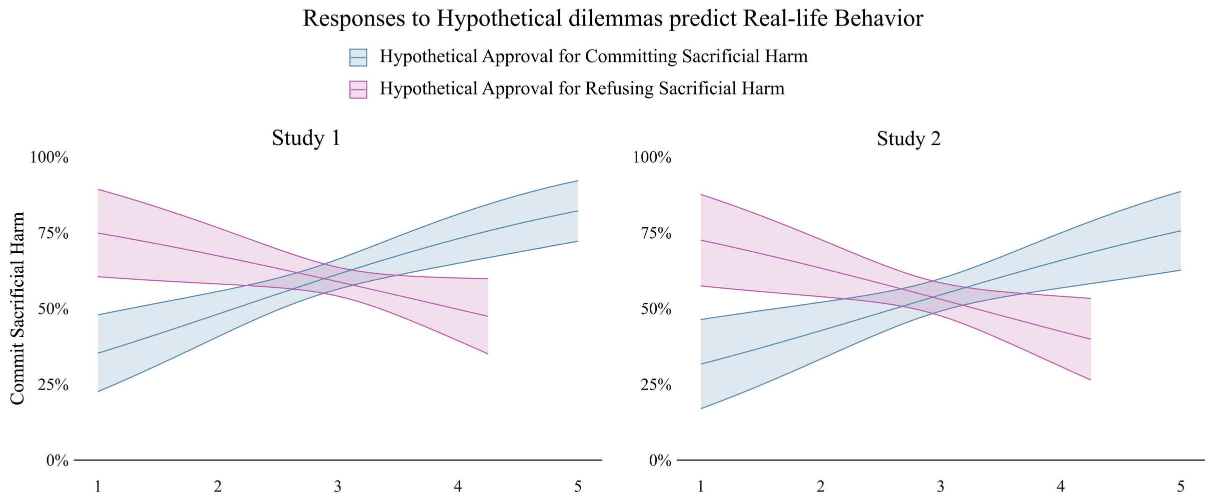
We also examined whether the gender composition of potential targets (dummy-coded) affected participants' decisions in Study 2. A logistic regression analysis showed that in the male-minority condition (where redirecting the shock harms a male target), 57% of participants chose to commit sacrificial harm. In the female-minority condition (where redirecting the shock harms a female target), only 51% of participants chose to commit sacrificial harm. However, this difference was not statistically significant, $\hat{b} = -0.27$, $z = -1.25$, $p = .212$. Interestingly, the approval measures in Study 2 suggested a more nuanced pattern. Approval ratings for committing sacrificial harm were lower in the female-minority condition compared with the male-minority condition, $\hat{b} = -0.29$, $t(353) = -2.17$, $p = .031$. For approval ratings of refusing sacrificial harm, no significant difference was uncovered, $\hat{b} = -0.08$, $t(353) = -0.71$, $p = .481$. Accordingly, while gender composition did not significantly affect behavioral choices, it did appear to modestly influence participants' evaluations of committing sacrificial harm, with participants showing less approval for actively harming a female target to save two male targets from harm compared to the opposite situation.

Prediction by Hypothetical Dilemmas

We next investigated whether responses to the hypothetical sacrificial dilemmas (in the online presurvey) predicted real-life behavior. We averaged participants' appropriateness ratings across the four hypothetical dilemmas for each response type (committing vs. refusing sacrificial harm), then entered both scores as simultaneous predictors in a logistic regression model with real-life decision as the outcome variable. This analysis on the Study 1 data revealed that higher appropriateness ratings for sacrificial harm (averaged across all hypothetical dilemmas) predicted corresponding behavior in real life ($\hat{b} = 0.53$, $z = 3.57$, $p < .001$). Higher appropriateness ratings for refusing sacrificial harm had the opposite effect, $\hat{b} = -0.37$, $z = -1.96$, $p = .050$ (see Figure 1). The associated Nagelkerke's R^2 of .06 suggests a small to moderate amount of variability is captured by these combined predictors. We found stronger effects on the approval measures of the shock options: Hypothetical approval for committing sacrificial harm correlated significantly with real-life approval ratings for the same action ($r = .36$, $p < .001$); and similarly, hypothetical approval for refusing sacrificial harm correlated with real-life approval ratings for refusing harm ($r = .30$, $p < .001$).

We replicated these effects in Study 2. A logistic regression model showed that hypothetical appropriateness ratings for committing sacrificial harm positively predicted corresponding behavior in the real-life dilemma ($\hat{b} = 0.48$, $z = 2.84$, $p = .005$), while appropriateness ratings for refusing sacrificial harm negatively predicted sacrificial choices ($\hat{b} = -0.43$, $z = -2.18$, $p = .029$). As in Study 1, both variables jointly predicted a small to moderate amount

Figure 1
Probability of Committing Harm in the Real-Life Dilemma Based on Responses to the Hypothetical Dilemmas



Note. See the online article for the color version of this figure.

of behavioral variance, Nagelkerke's $R^2 = .06$. The pattern of correlations between hypothetical and real-life approval ratings was also replicated, with hypothetical approval for committing sacrificial harm correlating with real-life approval for this action ($r = .30$, $p < .001$) and hypothetical approval for refusing harm correlating with real-life approval for refusing harm ($r = .19$, $p < .001$).

Motivations

To better understand why participants chose to act in a certain way, we analyzed the motivations they had written down after they had enacted their decision by categorizing these motivations. Some participants motivated their decision in multiple ways. If so, we tagged these motivations as belonging to multiple categories. We chose to create categories that were as close as possible to the written motivations of participants to preserve the diversity within the data. To visualize the motivational patterns associated with each response, we created Euler plots. These plots show the proportional frequency with which each motivation was mentioned and display the overlap whenever multiple motivations were mentioned. Motivations that occurred in fewer than 2% of instances in both studies are not included in the plots provided here. Table 1 provides a list of the categories we distilled from the responses. Figures 2 and 3 show the types of motivations mentioned by participants for their initial decisions.

Across both studies, a large majority of participants who decided to commit sacrificial harm motivated their judgment by referring to a cost–benefit analysis in line with utilitarian morality (see Figure 2). For instance, one participant wrote, "Either two people would get shocked or only one, I chose the smaller group." While this type of motivations dominated in both studies, a key difference emerged for the frequency and nature of the motivations expressing a preference for a specific target. In Study 1, such motivations were relatively rare and typically based on subjective impressions of the targets: "Intuition. The single person looked like they deserved it." In contrast, such motivations were much more present in Study 2, with gender emerging as a key motivational influence. Many participants explicitly expressed reluctance to harm women: "[I would rather have that one person receives a shock instead of two]. I prefer to 'rescue' girls from receiving shocks," but a smaller number of participants expressed the opposite preference: "Women have a higher pain threshold [and you're only hurting one person instead of two]." Beyond these target-specific considerations, some participants in both studies wanted to balance out judgments they believed others would make, "I assumed most people would not shock the single person and would just walk away. Therefore, I chose to interfere." Overlapping categories occurred when participants noted multiple motivations alongside their primary motivation. For instance, a

Table 1
List of Motivation Categories Compiled From the Participants' Written Statements About Their Real-Life Dilemma Decisions

Category	Description
Cost–benefit	Indicates it is better to harm one person than it is to harm two people.
Equal right	Indicates all targets have an equal right not to be shocked.
No harm	Indicates they do not want to harm others.
Inaction	Indicates it is better to do nothing.
Button	Indicates difficulty with pressing the button.
No responsibility	Indicates that they do not want to be responsible for the outcome.
Fate	Indicates that the outcome has already been determined, potentially by an external force such as god or fate.
Preferred victim	Indicates a preference for a specific target.
Shared pain	Indicates that having to suffer alone is worse than suffering in group.
Balance harm	Indicates a wish to balance out the (assumed) choice of other participants or, on the second iteration, their own prior choice.
Vibes	Indicates their choice was driven by feeling or intuition.

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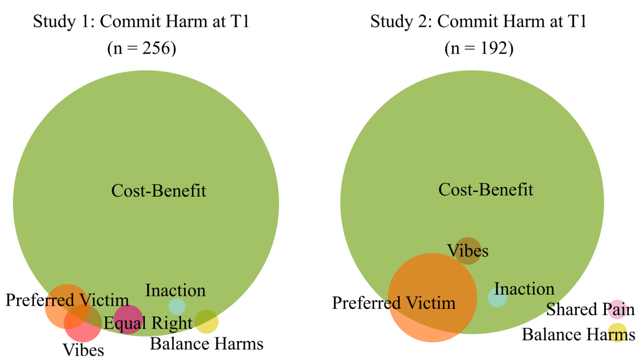
Figure 2

Euler Plots of Participants' Motivations When Committing Sacrificial Harm



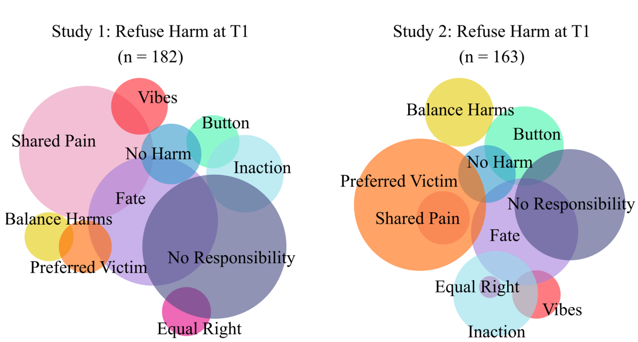
Note. T1 = Time 1. See the online article for the color version of this figure.

motivation that contains both elements of the "Vibes"-category and a cost–benefit analysis: "I found it more acceptable to hurt one person than several. In my head, I feel bit less guilty."

Interestingly, across both studies, participants who refused to commit the sacrificial harm expressed a much larger variety of motivations (see Figure 3). Some participants expressed unease about harming others "I don't think you should change what happens if that harms someone", a motivation that one might argue is deontological in nature as it directly refers to a refusal to harm others. Some participants emphasized equal moral status, arguing that all potential victims had equal rights not to be harmed: "C is not less worthy than A & B, it would be different if C was a loved one". Other motivations seemed to reflect a refusal to make a decision. For instance, one participant wrote, "Fate has decided who was sitting where and I did not want to be responsible for who was the victim," a motivation that touches on both the fate and no responsibility categories. Some participants expressed unease with pressing the button: "It felt harder to push the button than to do nothing" or unease with acting: "I couldn't choose so then I'd rather do nothing than something". Remarkably, many participants suggested that pain experienced in a group is preferable over pain experienced alone: "The single person was already unlucky to be sitting alone.

Figure 3

Euler Plots of Participants' Motivations When Refusing Sacrificial Harm



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In the pair of people, there are two, so they have each other [for solace]". Similar to our findings for committing sacrificial harm in Study 2, also for refusal to commit harm, there was notable increase in target preferences based on gender.

Second Iteration

Next, we analyzed how participants responded to the second dilemma iteration. On the one hand, this iteration involves the same targets and the same type of decision (i.e., "Should you shock one to save two?"). Accordingly, one might expect participants to respond consistently with their initial choice to maintain a sense of ethical steadfastness. On the other hand, this second iteration is contextualized by prior events. Participants are aware that some targets have already received a shock and might take this additional context into account when making their decision. This information could lead participants to alter their choice in an attempt to balance out the distribution of harm, similar to earlier findings in studies using hypothetical iterative dilemmas (Bostyn & Roets, 2022; Roets et al., 2020).

Switching Behavior

As expected, a substantial proportion of participants did switch their decisions across iterations: 34.9% in Study 1 and 23.5% in Study 2, thereby demonstrating how powerfully this context can shape moral choices even when the core dilemma structure remains unchanged. Note that the experiment was set up in such a way that the outcome of the first trial did not always reflect the decision that participants themselves had made. Due to the lottery nature of our study design, some participants saw their initial decision executed, while others did not. In Study 1, to test whether this influenced the likelihood of switching, we ran a 2 × 2 logistic regression with (a) participants' initial decision and (b) whether that decision was executed, as dummy-coded predictors. This model demonstrated a significant main effect of whether participants saw their decision executed ($\hat{b} = 1.71$, $z = 4.78$, $p < .001$), but no effect of the type of decision itself, or their interaction (both $|z| < 0.91$, both $p > .367$). Participants witnessing that their choice was not executed only switched in 14% of cases, whereas 54% of those who had seen it executed switched. In fact, for participants who saw their initial decision executed, the effect of context was so strong that a logistic regression showed no significant relationship between first and second decisions ($\hat{b} = -0.32$, $z = -1.18$, $p = .237$).

In Study 2, we conducted the same analyses and found similar but not identical results. Whether participants' initial decision was executed again significantly predicted switching behavior ($\hat{b} = 1.02$, $z = 2.35$, $p = .019$) and again no effect of initial decisions or an interaction effect emerged (both $|z| < 1.28$, both $p > .200$). Participants who did not see their initial decision executed switched in 11% of cases, compared with 35% of those who saw their decision executed. However, unlike Study 1, when examining only participants whose initial decision was executed, their first choices still significantly predicted their second choices ($\hat{b} = 1.37$, $z = 4.24$, $p < .001$). The stronger relationship between first and second choices in Study 2 may reflect gender-based preferences that remained consistent across iterations, with participants potentially making more principled decisions about which gender group should bear the burden of harm, a consideration absent in the gender-homogeneous groups of Study 1.

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Motivations

Finally, we analyzed the motivations for the second decision. We created Euler plots for participants' motivations split according to what participants' first decision was, whether that response was executed, and how they responded to the second iteration.

We first considered the motivations of participants who committed or refused the sacrificial harm, saw that decision executed, and decided in the same vein the second time (Figure 4).

Compared with the first iteration, we see no meaningful change in the types of motivations participants mention. The findings from Study 2 largely replicate those of Study 1, with participants who committed harm predominantly citing cost–benefit considerations and those who refused harm expressing a more diverse range of motivations. However, in Study 2, we observed an increased proportion of motivations suggesting a preference for a specific target.

Figure 5 shows the motivations for participants who made the same decision twice, but whose initial decision was not executed. Across both Studies 1 and 2, decisions to commit harm and decisions to refuse harm still show the same kinds of motivations as in Figures 2, 3, and 4. However, the motivation to "Balance Harms" is clearly more prominent in these subsets. This makes sense: All the reasons why participants favored their initial choice are still valid, and additionally, participants who care about how harms are balanced now also have a reason to favor that decision. As one participant who committed sacrificial harm notes, "Because otherwise, two people would have received a shock, now only a single person. Additionally, the two people had already been shocked."

Finally, Figure 6 shows the motivations of all those participants who favored different responses across iterations. While some other motivations are mentioned as well, it is clear that these are almost exclusively driven by concerns to balance harms. For participants who saw their initial decision executed (i.e., the largest groups), these decisions are logical: for example, "I felt it was bad to shock the two again because they had received the shock the previous time." There was however, also a small proportion of participants (3% and 3%) who switched even if they did not see their initial decision executed. What appears to have driven these judgments is that some participants balanced not what had actually occurred, but their own prior decision: "I consciously decided to do the opposite of what I had done before."

Discussion

Moral psychologists have extensively used sacrificial dilemmas to probe moral cognition and interpret responses to such dilemmas

Figure 4

Euler Plots of Participants' Motivations on the Second Dilemma Iteration for Participants Who Made the Same Decision Twice and Saw Their Initial Decision Executed

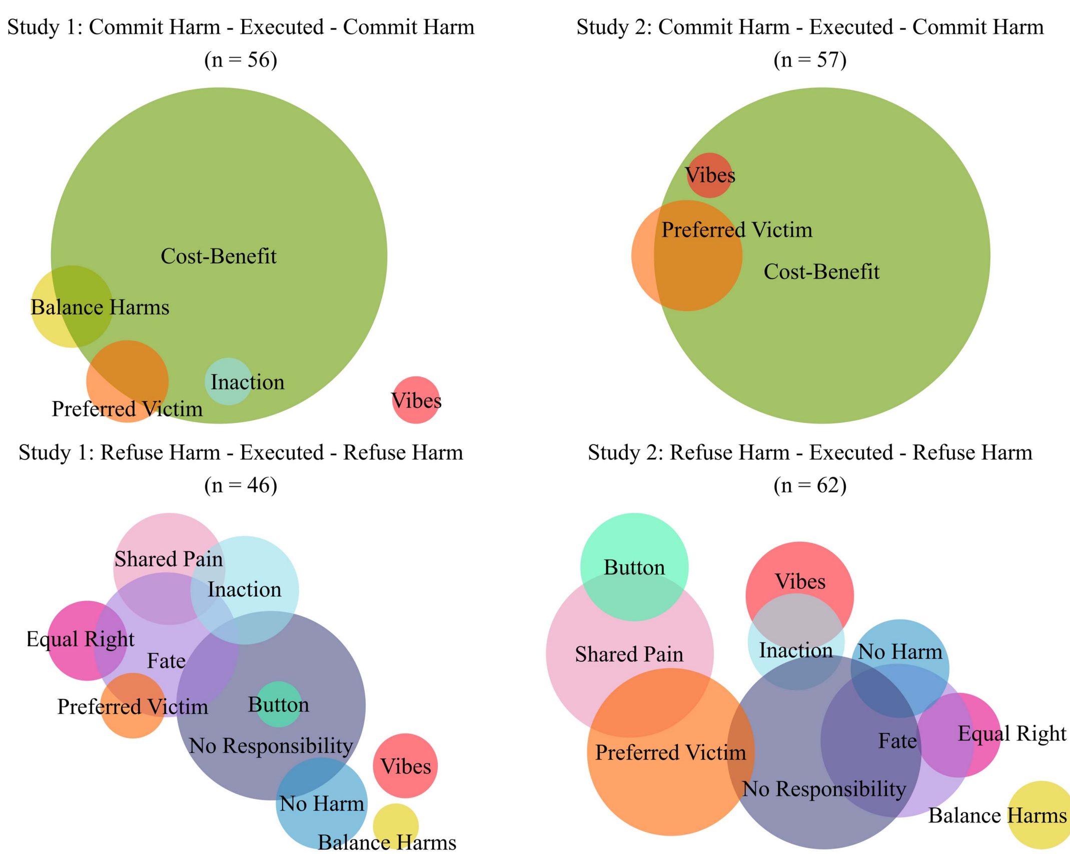

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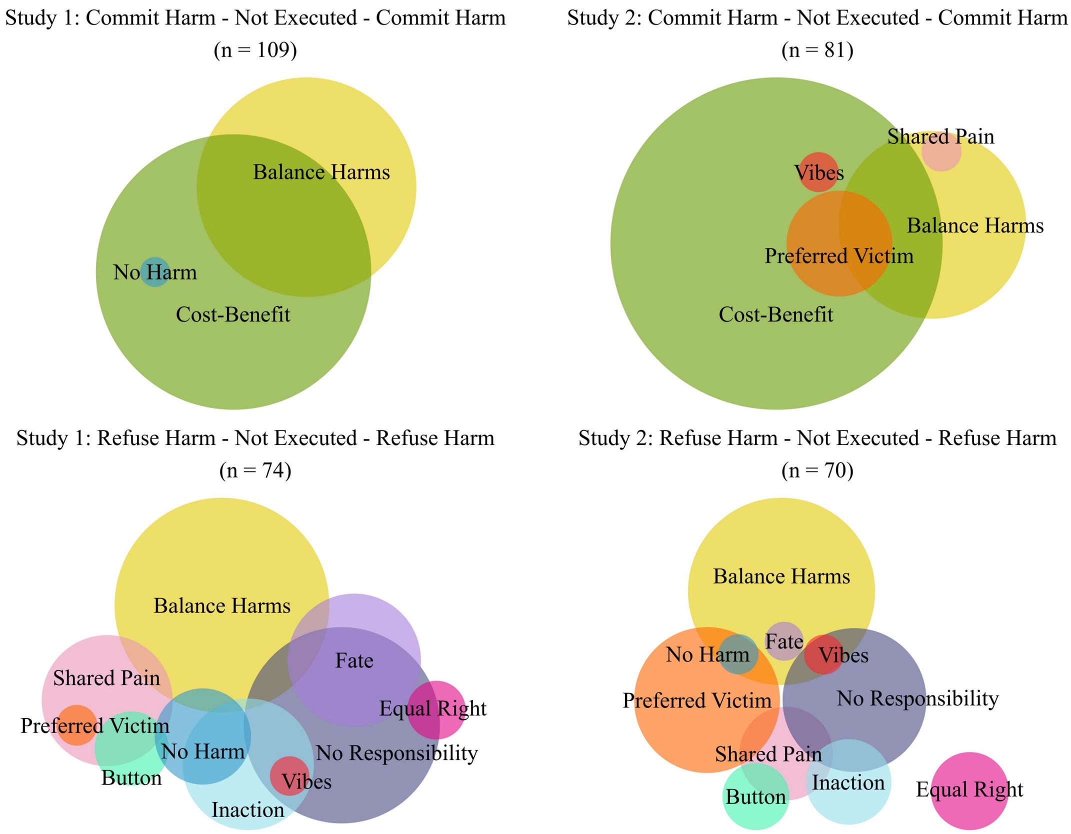
Figure 5

Euler Plots of Participants' Motivations on the Second Dilemma Iteration for Participants Who Made the Same Decision Twice, but Did Not See Their Decision Executed



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as reflecting either "utilitarian" or "deontological" moral concerns (Conway et al., 2018; Cushman & Greene, 2012; Greene et al., 2001). Nearly all research on such dilemmas has explored how people respond to hypothetical dilemmas. We developed a lab-based paradigm involving real physical harm. We aimed to advance our understanding of sacrificial decision making by (a) examining how contextual factors (physical proximity, target gender) influence moral decisions, (b) testing whether responses on traditional, hypothetical dilemmas predict real-life choices, (c) exploring how prior outcomes affect subsequent decisions, and (d) investigating motivations that emerge in consequential dilemmas.

First, our findings suggest that certain contextual factors may have less influence on moral decisions in real-life situations than traditional research with hypothetical dilemmas would predict. Although research on hypothetical dilemmas has consistently shown that people tend to refuse sacrificial harm when it requires up-close personal action (Greene et al., 2004, 2009; Tasso et al., 2017), our manipulation of physical proximity did not alter participants' decisions. Similarly, while prior research has shown gender-based preferences in moral judgments (Awad et al., 2018; FeldmanHall et al., 2016), the gender composition of potential targets produced only a minor effect on approval ratings and no effect on actual behavioral choices.

Several factors may explain these findings. First, in Study 1, our operationalization of the personal/impersonal distinction focused on physical proximity and touch, rather than "personal force," which some research suggests may be the more crucial element (Greene et al., 2009; though see also Francis et al., 2017). In Study 2, the minimal effect of gender composition on behavioral choices may be related to our predominantly female sample. A male-dominated sample might demonstrate different patterns of gender-based preferences than observed in our studies. Notably, participants' motivations revealed that gender was in fact frequently considered, but with conflicting directional preferences that may have canceled out an overall behavioral effect. More generally, the lottery design and delayed outcome in our paradigm might have diluted participants' sense of responsibility. While this was a conscious design decision implemented for ethical reasons (see the Ethics section), it might have also made our manipulations less impactful. Nevertheless, we still consider the limited effect of these manipulations surprising. Having to walk up to someone and press a button attached to their body while that person is fully aware of the decision being made arguably represents a qualitatively different emotional experience than making that same decision from behind a one-way mirror. Please note that other limitations are addressed in Table 2.

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Figure 6

Euler Plots of Participants' Motivations on the Second Dilemma Iteration for Participants Who Switched Their Decision Across Iterations

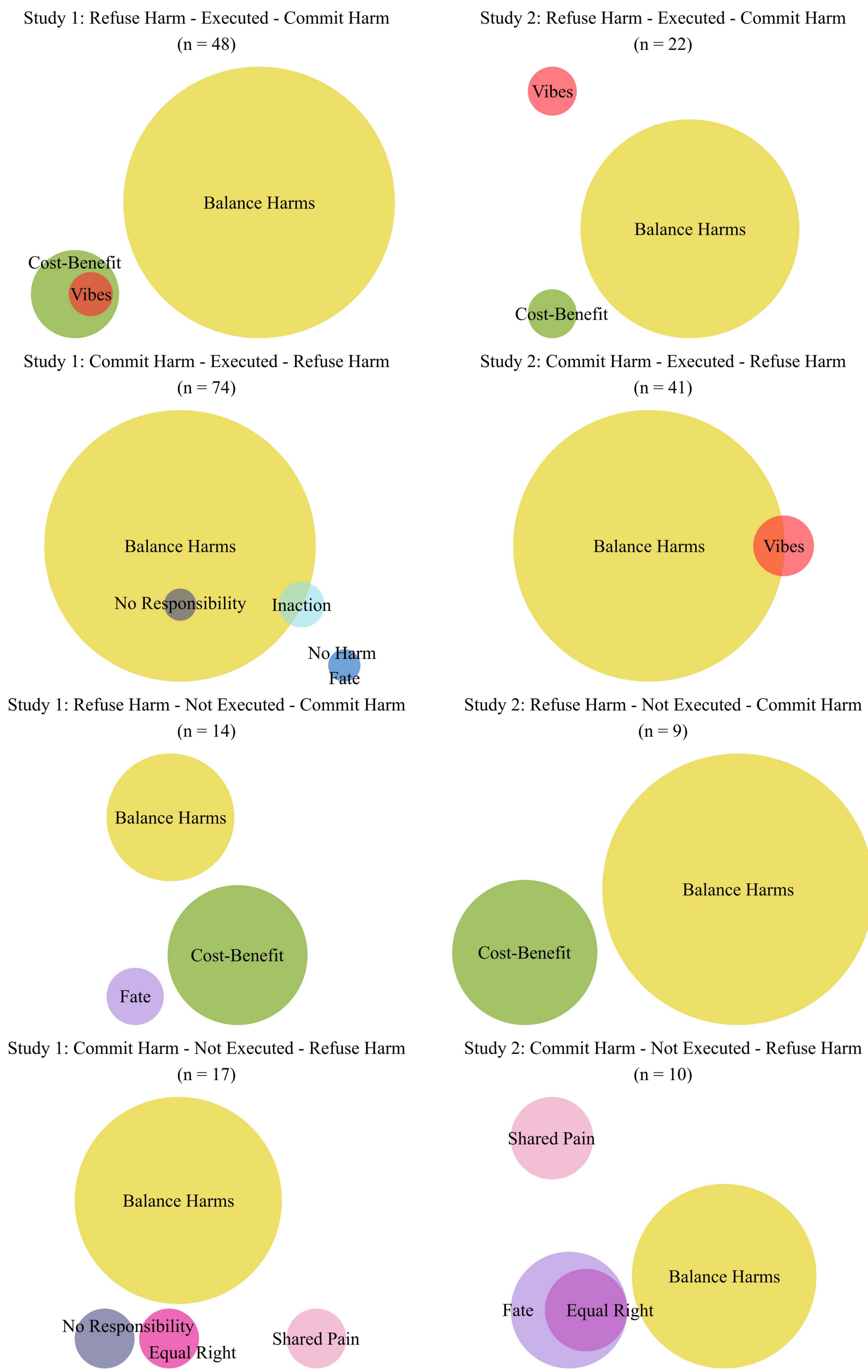

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Table 2
Table of Limitations

Dimension	Assessment
Internal validity	
Is the phenomenon diagnosed with experimental methods?	Yes
Is the phenomenon diagnosed with longitudinal methods	No
Were the manipulations validated with manipulation checks, pretest data, or outcome data?	The personal/impersonal manipulation's effectiveness was indirectly validated through the physical setup (same room vs. one-way mirror, button attached to target vs. in separate room). However, we did not include explicit manipulation checks measuring perceived personal involvement or psychological distance. Importantly, our operationalization focused primarily on physical proximity rather than personal force, meaning we did not manipulate all aspects of the personal/impersonal distinction identified in previous literature. The gender manipulation in Study 2 was similarly validated through the physical setup (visible gender of targets), but without explicit checks measuring how participants perceived this manipulation. Given our predominantly female sample, gender-based preferences may have manifested differently than in a more balanced sample.
What possible artifacts were ruled out	NA
Methodological constraints	
Lottery design	The lottery design (where only one randomly selected participant's decision determined the outcome) was required by our ethics committee. This design element was implemented specifically to minimize psychological distress by reducing participants' sense of direct responsibility for harm. While ethically necessary, this creates interpretive challenges as it distances our paradigm from fully consequential moral decisions where outcomes directly follow from choices. Additionally, it potentially reduces emotional engagement with the dilemma, which could mute effects that might be stronger in fully consequential contexts. Finally, it creates uncertainty about whether participants' decisions reflected how they would act if they knew with certainty their choice would be executed.
Stakes discrepancy	Our paradigm necessarily employed lower physical stakes (mild electroshocks) than the life-or-death scenarios typically featured in traditional hypothetical dilemmas. This creates a fundamental stakes asymmetry when comparing responses across these contexts. The predictive relationship we observed between hypothetical and real judgments might be stronger if the stakes were matched across contexts. However, it's worth noting that while traditional dilemmas involve higher stakes content (death versus mild pain), they represent lower psychological stakes for participants who face no consequences from their choices. Our paradigm may create higher psychological stakes through genuine responsibility for causing actual harm to real people, potentially offsetting the lower physical stakes to some degree.
Target characteristics	Despite randomizing target assignments across sessions and matching gender within sessions in Study 1, unmeasured characteristics of our targets may have influenced participants' decisions. Subtle variations in appearance, demeanor, or perceived vulnerability could have affected choices in ways difficult to control experimentally.
Statistical validity	
Was the statistical power at least 80%	The proportion test and the reported logistical regressions had >80% power for medium to small-sized effects.
Was the reliability of the dependent measure established in this publication or elsewhere in the literature?	The primary dependent measures in these Studies were behavioral choices in the dilemma and participants' written motivations for these choices. We established reliability in coding motivations through independent coding by two researchers with a high interrater agreement (overall agreement > .98 for T1 and T2 motivations, respectively).
If covariates are used, have the researchers ensured they are not affected by the experimental manipulation before including them in comparisons across experimental groups?	NA
Were the distributional properties of the variables examined and did the variables have sufficient variability to verify effects?	Yes
Generalizability to different methods	
Were different experimental manipulations used?	The personal/impersonal distinction was manipulated in a single way (physical proximity and button placement). The use of only one operationalization limits our ability to establish the robustness of this effect. In Study 2, we manipulated the gender composition of potential targets (male versus female).

(*table continues*)

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Table 2 (*continued*)

Dimension	Assessment
Generalizability to field settings	
Was the phenomenon assessed in a field setting?	No
Are the methods artificial?	The laboratory setting introduces some artificiality, but our paradigm used real physical consequences (actual electroshocks) and real human targets, making it more ecologically valid than traditional hypothetical dilemma research. Nevertheless, important artificialities remain. First, the lab context lacks the rich embedded social context of real-world moral decisions, including potential preexisting relationships with those affected. Additionally, participants lacked reputational concerns that often accompany moral decisions outside the lab, as the group setting and random selection procedure provided plausible deniability. Our paradigm represents an intermediate step between purely hypothetical scenarios and fully consequential real-world moral decisions.
Generalizability to times and populations	
Are the results generalizable to different years and historic periods?	This was not tested. The fundamental psychological processes we uncovered likely reflect basic aspects of human moral psychology that are relatively stable across time periods. However, specific attitudes toward harm and the acceptability of different moral considerations may vary across historical contexts.
Are the results generalizable across populations (e.g., different ages, cultures, or nationalities)?	Our sample was limited to predominantly undergraduate students from a single Western European university, with a substantial gender imbalance (84% female in Study 1, 83% female in Study 2). This gender skew is particularly relevant for interpreting Study 2, which manipulated the gender composition of potential targets. The observed target-gender effect patterns might differ significantly in a more gender-balanced sample. The specific balance of utilitarian, deontological, and other moral concerns might also vary substantially across cultures, age groups, or nationalities. We believe that the fundamental psychological processes we uncovered (e.g., the emergence of diverse motivations, the influence of prior outcomes) extend beyond this specific population studied but the precise patterns and proportions of choices would almost certainly differ in more diverse samples.
Theoretical limitations	
What are the main theoretical limitations?	The key theoretical limitation centers on how we conceptualize and measure moral motivations. While examining participants' spontaneous explanations provides valuable insights, this approach faces the fundamental challenge that participants may offer post hoc rationalizations rather than accessing the actual psychological processes that drove their decisions. Several patterns in our data suggest that motivations at least partially reflect meaningful psychological processes rather than mere rationalizations: such as the strong alignment between specific motivations and behavioral patterns, particularly the consistent association between 'Balance Harm' motivations and switching behavior. Nevertheless, we cannot definitively determine which motivations represent genuine decision drivers versus post hoc justifications. This limitation affects our ability to draw strong conclusions about the causal role of specific moral considerations in sacrificial decision-making. Future research using experimental manipulations of these proposed motivational factors would provide stronger evidence of their causal influence.

Note. NA = not applicable; T1 = Time 1; T2 = Time 2.

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While proximity and gender had limited impact, temporal context proved remarkably influential. This became apparent when examining participants' responses to the second dilemma iteration. Prior outcomes dramatically shaped subsequent moral judgments: 35% of participants in Study 1 altered their decision on the second iteration, with 23.5% switching in Study 2. Most striking was the pattern among participants who saw their initial choice executed: in Study 1, their second decision was completely unrelated to their first, indicating that consideration of prior outcomes entirely overrode initial moral preferences. In Study 2, while prior outcomes still strongly influenced second decisions, the initial choice remained a significant predictor, suggesting that here, gender-anchored decisions provided some resistance against equity-driven switching.

This sensitivity to prior outcomes highlights a crucial limitation of traditional sacrificial dilemma research. Traditional paradigms strip away temporal context by relying on single-shot, life-or-death scenarios. This is particularly problematic given that researchers often justify studying sacrificial dilemmas by suggesting that they reflect trade-offs people face in their daily lives. Real-life moral decisions mostly involve nonlethal outcomes within ongoing sequences often affecting the same individuals over time. Decision-makers must consider immediate trade-offs alongside past outcomes and future implications. This was reflected in participants' motivations, with many participants referencing a need to balance outcomes across iterations to ensure a fair distribution of harm over time. These considerations align with research showing that people often prioritize equality in resource

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allocation even when it conflicts with maximizing immediate welfare (Dawes et al., 2007). In the morality of sacrificial harm, fairness appears as critically important as utilitarian and deontological concerns (Bostyn & Roets, 2022; Roets et al., 2020).

Nuanced results also emerged when examining the relationship between hypothetical judgments and real-life behavior. Our findings demonstrate that responses to hypothetical dilemmas can predict real-life behavior, countering suggestions that such fantastical scenarios inevitably elicit unrealistic responses (Bauman et al., 2014; Davis, 2012; Elster, 2011; Körner et al., 2019). This provides important evidence that hypothetical dilemmas, despite their abstract and uncommon nature, tap into (some of) the underlying moral processes that guide real-life behavior. Participants at the opposite ends of hypothetical approval scales also showed dramatically different real-life behavior: Those who strongly approved of sacrificial harm in hypothetical scenarios were approximately 2.5 times more likely to commit such harm in our real-life paradigm compared to those who strongly disapproved. Nevertheless, the modest explained variance (Nagelkerke's $R^2 = .06$) indicates many real-life decision processes are not captured by hypothetical responses, echoing work showing structural differences between hypothetical and realistic moral decisions (Bostyn et al., 2018; Francis et al., 2016, 2017; Patil et al., 2014).

Our motivational analysis suggests this might reflect decisions being driven by concerns extending beyond traditional "utilitarian" and "deontological" frameworks. Despite using a sacrificial dilemma in its most basic form: "harm one to save two", participants expressed a rich spectrum of moral considerations. Participants who committed sacrificial harm predominantly expressed cost–benefit reasoning that aligned with utilitarian philosophy. However, among those who refused harm, explicit references to not wanting to harm others (the motivation most closely aligned with deontological principles) were only one of many diverse explanations offered. Some participants motivated their judgment by expressing a preference for a specific target. Others claimed that pain experienced as part of a group is less aversive than the same pain experienced by an individual. We also found participants who refused to accept moral responsibility for the outcome altogether, arguing that accepting luck-based "fate" was preferable over active human interference.

One could argue this pattern supports dual-process accounts of moral judgment as participants making the "utilitarian" choice consistently cited cost–benefit reasoning (potentially reflecting deliberative processes), whereas those refusing harm offered diverse explanations (potentially reflecting post hoc justifications for intuitive response, Cushman et al., 2006; Greene & Haidt, 2002; Haidt, 2001). However, a closer examination of our data challenges this interpretation. Most striking is the consistent association between "balance harm" motivations and switching behavior. Participants who switched decisions consistently cited fairness considerations regardless of the direction of their initial choice. If these motivations were post hoc rationalizations, switching participants should generate different justifications depending on their initial moral preference. Instead, we observed remarkable consistency in fairness-based reasoning. Some participants even switched despite their initial choice not being executed, arguing they wanted to balance their prior decision rather than actual outcomes. This is explicable from a fairness perspective but somewhat baffling for dual-process accounts.

These findings suggest that equity and fairness considerations might represent a significant factor in sacrificial moral decision making that traditional dual-process frameworks have overlooked. In our view, the emergence of fairness motivations should not be particularly controversial as fairness concerns have long been recognized as central to moral cognition (Fehr & Schmidt, 1999; Graham et al., 2011). Indeed, recent advances in computational moral psychology have also proposed frameworks going beyond the utilitarian–deontological dichotomy. Theories such as resource–rational contractualism (Levine et al., 2024) suggest that moral cognition involves psychological processes concerned with achieving mutually beneficial agreements, including fairness in distributing burdens and benefits.

For the remaining motivations, we do not mean to suggest that each of them represents a separate psychological mechanism, although we do believe that some of them might reflect meaningful differences in how people approach moral decisions. For instance, participants likely vary in the extent to which they feel obligated to act in dilemma situations (Tomasello, 2020), which appears reflected in the "No Responsibility" and "Fate" motivations, that is, "It's not my place to decide who gets shocked." Similarly, the "Shared Pain" motivation, the idea that "pain is more tolerable when experienced together," might relate to differences in how people weigh social support during adversity (Schachter, 1959). Our current data cannot establish that such considerations causally impacted judgments, but still suggest a richer array of psychological processes than traditional frameworks acknowledge. Future studies could experimentally manipulate factors related to these specific motivations. For instance, the "shared pain" motivation could be manipulated by increasing physical or psychological distance between the targets to disrupt the perception of a shared experience.

Conclusion

Some researchers have argued that sacrificial dilemmas focus on an abstract moral contrast that is disconnected from many of the problems we face in our everyday lives and are thus unsuitable for studying moral cognition (Bloom, 2011; Hester & Gray, 2020). This concern has merit, but the issue may lie not with sacrificial dilemmas themselves but with framing responses within an abstract philosophical dichotomy. Our work demonstrates that sacrificial dilemmas can be valuable tools for studying moral psychology when implemented in ways that capture real consequences and allow for a nuanced analysis of decision making. The findings not only validate a method for studying consequential sacrificial decisions in the lab but also reveal several promising avenues for future research. In particular, the emergence of fairness-based reasoning alongside utilitarian and deontological concerns raises questions about how people integrate equity considerations with other competing moral principles. To be clear, we do not intend to suggest that responses to sacrificial dilemmas cannot be meaningfully interpreted through utilitarian and deontological lenses. Yet, constraining our analyses to these perspectives may not do justice to the richness of human moral reasoning. By incorporating repeated choices and motivational analysis into sacrificial dilemma designs, researchers can transcend the utilitarianism–deontology dichotomy. Doing so will enhance our understanding of the variety of moral concerns at play and might reveal significant overlap in how people approach sacrificial dilemmas and how they deal with other types of moral conflict. Such a shift will not only lead to more robust and nuanced theoretical models of moral decision making but will enhance our ability to understand real-world moral choices.

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Received July 2, 2025
Accepted July 7, 2025 ■

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